Physics-Guided Temporal Signatures for Data-Scarce Temporal Event Recognition

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Abstract

End-to-end sequence models dominate temporal recognition when discriminative structure must be learned from abundant data. We study the complementary regime, in which the event mechanism is known a priori and labelled data are scarce, and formalise a recipe we call *physics-guided temporal signatures* (PGTS): name task primitives from the event mechanism, apply a fixed bank of temporal operators including an order-sensitive asymmetry statistic, select the single split parameter τ on training data only, and train a lightweight classifier. We give three theoretical results: a Bayes-optimal rule and error for the asymmetry statistic under an empirically supported heteroscedastic Gaussian model; an impossibility theorem showing any order-invariant feature map is exactly at chance on permutation-paired temporal-order constructions; and a new finite-sample theorem showing that thresholding the asymmetry statistic separates three-phase events from permuted negatives with failure probability decaying exponentially in sequence length. A controlled benchmark family with three constructions (three-phase falls, sit-to-stand ramps, direction-of-time reversal) confirms the theory: structured signatures and order-sensitive baselines solve all three, order-invariant features sit exactly at 0.500, and sweeps over noise, event timing, duration, length, and distractor channels chart where the inductive bias degrades. As the primary real-data case study, the Dynamic Trajectory Signature (DTS) instantiates PGTS for skeleton-based fall detection: under a strict twin-free protocol, DTS+HGB attains AUROC 0.9895 [0.9851, 0.9935] on held-out FallVision, best of thirteen evaluated models, significantly ahead of LSTM/GRU/Transformer, two ST-GCNs, MiniROCKET, and a TCN, and matching the strongest baseline, MultiROCKET (0.9868; paired margin not significant), with 388× fewer features. It leads at every training size, dominates MiniROCKET at every fixed false-positive rate (recall 0.941 vs 0.861 at 5% FPR), transfers zero-shot to URFD (0.9200) and GMDCSA-24 (0.9633), matches MultiROCKET under true leave-one-subject-out evaluation (0.941 vs 0.934; not significant at four subjects), and recovers the hard Bed fold faster than a matched MiniROCKET baseline at every few-shot budget. All artefacts, manifests, score vectors, seeded benchmark code, and a numbers audit are released.

1 Introduction

A central question in machine learning is when task-specific inductive bias, encoded as fixed features, can replace the representational power of learned sequence models. For temporal recognition at scale the answer has been settled in favour of end-to-end learning: spatial-temporal graph networks (Yan, Xiong, and Lin, 2018; Shi et al., 2019; Liu et al., 2020; Chen et al., 2021; Lee et al., 2023) and pretrained pose backbones (Duan et al., 2022) dominate large multi-class skeleton benchmarks, and convolutional transforms (Dempster, Petitjean, and Webb, 2020; Dempster, Schmidt, and Webb, 2021; Tan et al., 2022) and deep architectures (Ismail Fawaz et al., 2020; Bai, Kolter, and Koltun,

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2018) dominate generic time-series classification. That dominance rests on two conditions: the discriminative structure is unknown and must be learned, and enough labelled data exist to regularise high-capacity models.

This paper studies the complementary regime, common in safety-critical event detection, where both conditions fail: the event mechanism is formally known, and labelled data are scarce. Our claim is not that a particular application can be solved with handcrafted features; it is a testable claim about inductive bias in data-scarce temporal learning: *when event dynamics are known, a fixed temporal-signature map can remove the burden of learning temporal structure from a generic sequence model*. We formalise the recipe as *physics-guided temporal signatures* (PGTS): (1) name task primitives from the event mechanism; (2) apply a fixed bank of temporal operators, including a temporal-asymmetry statistic $\alpha(z;\tau)$, the fraction of a primitive's total path variation concentrated in the first τ of the sequence; (3) select τ on training data only; (4) train a lightweight classifier; (5) validate under low-data and domain-shift protocols.

We test this claim along three axes. **Theory.** Theorem 1 derives the Bayes-optimal rule and error for α under an empirically supported heteroscedastic Gaussian model. Theorem 2 proves that any order-invariant feature map is exactly at chance on permutation-paired temporal-order constructions, making temporal ordering provably necessary. Theorem 3 (new relative to prior drafts) upgrades an asymptotic separation argument to a finite-sample guarantee: thresholding α separates three-phase events from permuted negatives with failure probability $\leq 5\exp(-n\bar{\mu}^2\Delta^2/72B^2)$, with an explicit sample-length condition and a multi-primitive corollary. **Controlled benchmark family.** Three synthetic constructions, three-phase falls, sit-to-stand ramps, and a direction-of-time reversal task, make class-conditional frame multisets identical so only ordering differs; we sweep training-set size, drop SNR, event onset, event duration, sequence length, distractor channels, and operator subsets, comparing the structured signature against order-invariant features and generic order-sensitive learners. **Real-data case study.** The Dynamic Trajectory Signature (DTS), PGTS instantiated for skeleton-based fall detection with eight kinematic primitives and 16 operators (128 features), is evaluated under a strict leakage-corrected protocol against twelve baselines, with low-data learning curves, fixed-false-positive-rate operation, leave-session-out and leave-fall-origin folds, zero-shot transfer to two external datasets, true leave-one-subject-out validation, and an interpretable failure-and-recovery analysis of the hardest fold including a matched-protocol few-shot baseline comparison.

Contributions. (i) The PGTS formalisation and an explicit account of when it should and should not work (Section 6); (ii) three theoretical results, including a new finite-sample separation theorem with explicit constants and a matching impossibility theorem; (iii) an expanded controlled benchmark family in which the theory's predictions are exactly realised and the boundary of the inductive bias is charted empirically; (iv) a comprehensive, leakage-corrected fall-detection case study in which the 128-dimensional signature with gradient-boosted trees outperforms or matches LSTM/GRU/Transformer, two ST-GCNs, a TCN, MiniROCKET, and MultiROCKET under iid, low-data, fixed-FPR, session, fall-origin, cross-dataset, subject-held-out, and few-shot-adaptation protocols; and (v) a complete reproducibility package with saved score vectors, split manifests, a numbers audit, and seeded benchmark code.

Scope. Real-data superiority claims cover the evaluated baselines only: PoseC3D and other pretrained systems are out of scope because pretraining reintroduces the large-external-data condition our study isolates. FallVision subject IDs are not public, so leave-session-out is used there only as a subject proxy; true leave-one-subject-out evaluation is reported separately on GMDCSA-24, which publishes subject identifiers. Reported runtime covers signature extraction and inference only.

2 Related work

Skeleton-based action recognition and fall detection. ST-GCN (Yan, Xiong, and Lin, 2018) and successors (Shi et al., 2019; Liu et al., 2020; Chen et al., 2021; Lee et al., 2023; Duan et al., 2022) dominate large multi-class skeleton benchmarks; pretrained PoseConv3D-style backbones increasingly support few-shot fine-tuning. PGTS targets the complementary binary, data-scarce regime. Pose-based fall detectors trained on URFD reach 97–98% within-dataset (Lin and Dong, 2021; Zhu et al., 2025); our URFD evaluation is zero-shot transfer. Surveys document device and environment shift as persistent failure modes (Igual, Medrano, and Plaza, 2013; Mubashir, Shao, and Seed, 2013; de Miguel et al., 2017).

Time-series classification. ROCKET/MiniROCKET/MultiROCKET (Dempster, Petitjean, and Webb, 2020; Dempster, Schmidt, and Webb, 2021; Tan et al., 2022) are strong generic transforms; InceptionTime (Ismail Fawaz et al., 2020) and TCNs (Bai, Kolter, and Koltun, 2018) are standard deep baselines. We evaluate MiniROCKET, MultiROCKET, and a TCN on the main split, MiniROCKET on the controlled benchmark and under few-shot Bed adaptation, and MultiROCKET under subject-held-out transfer; InceptionTime is released as a configured script.

Inductive bias, sample efficiency, and operating points. Structured inductive biases are a recurring route to sample efficiency (Battaglia et al., 2018), with task-specific features helping in data-scarce medical signal processing (Rajpurkar et al., 2017) and anomaly detection (Pang et al., 2021). Distribution shift and the gap between ranking and calibrated operating points (Guo et al., 2017; Gulrajani and Lopez-Paz, 2021) motivate our fixed-FPR, transfer, and threshold-shift analyses.

3 Theory: asymmetry, impossibility, and finite-sample separation

Definition 1 (Temporal asymmetry operator)**.** *For a scalar primitive trajectory $z = z_{1:T}$ and split $\tau \in (0, 1)$, let $\Delta z_t = z_t - z_{t-1}$ and*

$$\alpha(z;\tau) = \frac{\sum_{t=2}^{\lfloor \tau T \rfloor} |\Delta z_t|}{\sum_{t=2}^{T} |\Delta z_t| + \epsilon}, \qquad \epsilon = 10^{-8}. \tag{1}$$

α is the fraction of total path variation concentrated in the first τ of the sequence: events that concentrate motion late (falls) have small α; permuted or temporally diffuse signals have $\alpha \approx \tau$.

Theorem 1 (QDA Bayes bound for temporal asymmetry)**.** *Let $\alpha \mid c \sim \mathcal{N}(\mu_c, \sigma_c^2)$, $c \in \{f, n\}$, with equal priors and $\sigma_f \neq \sigma_n$. The Bayes-optimal classifier is a quadratic discriminant rule with two real crossover points $x_1 < x_2$ solving*

$$\Big(\tfrac{1}{\sigma_f^2} - \tfrac{1}{\sigma_n^2}\Big)x^2 - 2\Big(\tfrac{\mu_f}{\sigma_f^2} - \tfrac{\mu_n}{\sigma_n^2}\Big)x + \Big(\tfrac{\mu_f^2}{\sigma_f^2} - \tfrac{\mu_n^2}{\sigma_n^2}\Big) - 2\ln\tfrac{\sigma_n}{\sigma_f} = 0, \tag{2}$$

with fall assigned to the interior (x_1, x_2) when $\sigma_f < \sigma_n$, and Bayes error $\varepsilon^*_{\text{QDA}} = \frac{1}{2}(P(\alpha \notin (x_1, x_2) \mid f) + P(\alpha \in (x_1, x_2) \mid n))$.

On FallVision's hip-speed primitive at $\tau^* = 0.30$, $\hat{\mu}_f = 0.171$, $\hat{\sigma}_f = 0.165$, $\hat{\mu}_n = 0.381$, $\hat{\sigma}_n = 0.220$, and Levene's test rejects equal variance ($p = 1.99\times10^{-24}$); (2) gives $x_1 \approx -0.50$ (outside the support of α, so a single threshold $x_2 \approx 0.308$ is active), $P(\text{miss} \mid f) = 0.202$, $P(\text{fa} \mid n) = 0.369$, and $\varepsilon^*_{\text{QDA}} = 0.286$. The full eight-primitive signature reduces observed held-out error to 0.051, a factor-5.6 reduction over this single-primitive bound.

Theorem 2 (Order-invariance impossibility)**.** *Call ϕ order-invariant if $\phi(x_{\sigma(1)}, \ldots, x_{\sigma(T)}) = \phi(x)$ for every $\sigma \in S_T$. In any construction where each negative is $X^- = \sigma(X^+)$ with $X^+ \sim P_+$ and σ a (random or deterministic) permutation independent of X^+, every classifier $g = h \circ \phi$ on an order-invariant ϕ has identical class-conditional score laws; hence the ROC curve is the diagonal, $\text{AUROC} = \frac{1}{2}$, and the Bayes error is $\frac{1}{2}$.*

Theorem 3 (Finite-sample asymmetry separation)**.** *Model an event primitive by n independent increment magnitudes $w_t \in [0, B]$ with mean μ_s on a still phase $t/n \in [0, p_1)$ and floor phase $[p_2, 1]$, and mean $\mu_d \geq \mu_s$ on the drop phase $[p_1, p_2)$; let the matched negative be a uniformly random permutation of the same increments. Fix $\tau \leq p_1$; set $\bar{\mu} = (p_1{+}1{-}p_2)\mu_s + (p_2{-}p_1)\mu_d$, $a_f = \tau\mu_s/\bar{\mu}$, and $\Delta = \tau - a_f = \tau(p_2{-}p_1)(\mu_d{-}\mu_s)/\bar{\mu} > 0$. Then for all $n \geq 6/\Delta$, the rule "event iff $\alpha \leq \theta$" with $\theta = a_f + \Delta/2$ satisfies*

$$P(\alpha_{\text{event}} \geq \theta) + P(\alpha_{\text{neg}} \leq \theta) \;\leq\; 5\exp\Big(-\frac{n\,\bar{\mu}^2\Delta^2}{72\,B^2}\Big), \tag{3}$$

hence $\text{AUROC}(\alpha) \geq 1 - 5\exp(-n\bar{\mu}^2\Delta^2/72B^2)$ and balanced error $\leq \frac{5}{2}\exp(-n\bar{\mu}^2\Delta^2/72B^2)$.

Corollary 1 (Multiple primitives)**.** *If k primitives independently satisfy Theorem 3 with individual failure probabilities $q < \frac{1}{2}$, majority voting over the single-primitive rules has error at most $\exp(-2k(\frac{1}{2} - q)^2)$.*

Model	Fall		Sit-to-stand		Reversal	
	n=100	n=1000	n=100	n=1000	n=100	n=1000
DTS+HGB (128-d)	0.995±0.002	1.000	0.998±0.002	1.000	1.000	1.000
α-only+LR (8-d)	1.000	1.000	1.000	1.000	1.000	1.000
MiniROCKET+Ridge	1.000	1.000	1.000	1.000	1.000	1.000
Raw-seq MLP	1.000	1.000	1.000	1.000	0.993±0.010	1.000
Bag-of-frames+HGB	0.500	0.500	0.500	0.500	0.500	0.500

Table 1: Expanded controlled temporal-order benchmark: held-out AUROC (mean±sd over five seeds, 500 test pairs) on three constructions in which negatives are frame permutations (Fall, Sit-to-stand) or exact time reversals (Reversal) of positives, so class-conditional frame multisets are identical and Theorem 2 applies. Order-invariant bag-of-frames is exactly at chance everywhere; the structured signature and order-sensitive baselines solve all three constructions.

Proofs are in Appendix A; the negative side uses Hoeffding–Serfling concentration for sampling without replacement (Hoeffding, 1963; Serfling, 1974), and the impossibility proof is pointwise in σ, covering deterministic reversal as well as uniform permutation. Three remarks. *(a)* Theorem 3 makes the sample-length condition explicit: error $\leq \delta$ once $n \geq \frac{72B^2}{\bar{\mu}^2\Delta^2} \ln \frac{5}{\delta}$, quantifying when a single fixed statistic suffices and no sequence learning is needed. *(b)* Constants are not optimised; the released verification script confirms the bound and shows empirical failure decaying exponentially with far better constants (e.g., the left side of (3) falls from 0.192 at n=60 to 0.003 at n=500 at the parameters in Appendix A). *(c)* A crude capacity heuristic comparing $d = 128$ fixed features to a 216k-parameter LSTM ($n^* = d_{\mathrm{LSTM}}/\ln d_{\mathrm{LSTM}} \approx$ 17,599, Appendix B) is reported as intuition only; the low-data claim is supported empirically by the learning curves below, not by the heuristic.

4 Controlled temporal-order benchmark family

The benchmark is synthetic by design: it realises the assumptions of Theorems 2 and 3 exactly, so the theory's predictions can be checked rather than assumed. Three constructions make class-conditional frame multisets identical so that only temporal ordering carries label information: *fall* (three-phase trajectories: still, drop, floor; negatives are uniform frame permutations), *sit-to-stand* (rise-and-plateau ramps; permuted negatives), and *reversal* (early-burst trajectories; negatives are exact time reversals, isolating direction-of-time information). Each construction uses eight informative channels, pair-level splits, 500 held-out test pairs, and τ selected on training data only by the summed-Welch rule; defaults are T=150 and drop SNR 8.

Table 1 reports the headline: DTS-style signatures with gradient boosting and α-only logistic regression solve all three constructions, generic order-sensitive learners (MiniROCKET, an MLP on raw sequences) also succeed, and the order-invariant bag-of-frames classifier is exactly at 0.500 everywhere, as Theorem 2 requires. Appendix Table 8 probes the boundary of the inductive bias with sweeps over drop SNR (1–16), event onset p_1 (0.15–0.7), event duration (0.05–0.4), sequence length (30–300), distractor channels (0–120), and operator subsets. Three findings: (i) order-invariant operator subsets of the signature itself (quantiles, min/max) sit exactly at chance, so the order-sensitive operators carry all discriminative signal; (ii) α-only degrades as SNR approaches 1 (0.879±0.004) and dips mildly under a misspecified fixed τ or very short events, while the full bank and train-selected τ remain at ceiling, consistent with the $\tau \leq p_1$ condition and SNR dependence of Theorem 3; and (iii) at these settings every order-sensitive method, structured or not, solves the constructions even with 120 distractor channels, so the family measures the necessity of temporal order, not separation among order-sensitive methods; the real-data case study below supplies that separation under data scarcity. The original single-construction result (signature AUROC 1.000 at n=100; α-only ≥ 0.996; bag-of-frames exactly at chance for every n) is reproduced in Appendix Figure 7.

5 Case study: data-scarce skeleton fall detection

Falls have a kinematically distinctive three-phase structure that is formally known rather than latent, and public training data are scarce, making fall detection the natural real-data instantiation of PGTS.

Model	AUROC	F1	Precision	Recall	Accuracy	FP	FN
LSTM	0.9558 [0.9446, 0.9662]	0.905	0.893	0.916	0.900	63	48
GRU	0.9712 [0.9623, 0.9795]	0.920	0.913	0.927	0.917	51	42
Transformer	0.9810 [0.9749, 0.9872]	0.940	0.947	0.934	0.939	30	38
Compact ST-GCN	0.9185 [0.9016, 0.9340]	0.849	0.851	0.847	0.845	85	88
FullST-GCN-COCO	0.9652 [0.9553, 0.9740]	0.907	0.895	0.920	0.903	62	46
MiniROCKET	0.9760 [0.9672, 0.9839]	0.936	0.931	0.941	0.934	40	34
MultiROCKET†	0.9868 [0.9808, 0.9920]	0.949	0.954	0.944	0.948	26	32
TCN†	0.9629 [0.9523, 0.9734]	0.910	0.893	0.929	0.906	64	41
DTS+LR	0.9433 [0.9297, 0.9559]	0.890	0.883	0.897	0.886	68	59
DTS+ET	0.9801 [0.9734, 0.9861]	0.936	0.907	0.967	0.932	57	19
DTS+RF	0.9785 [0.9711, 0.9849]	0.927	0.893	0.963	0.922	66	21
DTS+HGB	**0.9895** [0.9851, 0.9935]	**0.951**	0.934	0.969	**0.949**	39	18
DTS-Net	0.9775 [0.9699, 0.9849]	0.928	0.927	0.929	0.926	42	41

Table 2: FallVision held-out results, strict twin-free split (1,115 clips: 574 falls, 541 non-falls). AUROC carries 95% bootstrap intervals; 95% intervals for the remaining metrics are recorded in the released artefacts. Thresholds tuned on validation F1 for every model. †Baselines added on identically rebuilt sequences under the same validation-only protocol; harness fidelity: rebuilt DTS features reproduce DTS+HGB's 0.9895 exactly, and MiniROCKET re-run in this harness gives 0.9797, within its published interval.

The Dynamic Trajectory Signature. Skeleton sequences $S \in \mathbb{R}^{T\times 17\times 2}$ are hip-centred (midpoint of COCO joints 11–12) and torso-normalised (median shoulder-to-hip distance); missing keypoints are linearly interpolated and keypoints with confidence below 0.10 are zeroed. Eight primitives are extracted per clip (bounding-box width, hip height, torso angle, hip speed, centre-of-mass speed, hip acceleration, shoulder height, head height), each summarised by the 16-operator bank $\Omega(z;\tau) = [\mu, \sigma, \min, \max, Q_{25}, Q_{50}, Q_{75}, R, z_1, z_T, \delta, \beta, \sum |\Delta z|, |\Delta z|_{\max}, |\Delta^2 z|_{\max}, \alpha(z;\tau)]$, yielding $\phi(X) \in \mathbb{R}^{128}$. τ^* maximises the summed two-sample Welch statistic over primitives on the training split (13-point grid on $[0.20, 0.80]$); on FallVision $\tau^* = 0.30$, reused unchanged for external transfer. DTS-Net, an interpretable neural variant (asymmetry-guided attention over the eight blocks; 10,506 parameters), is exploratory; tree ensembles give peak accuracy.

Data, deduplication, and protocol. Parsing the FallVision keypoint archives (FallVision Dataset, 2024) yields 5,793 clips (2,966 falls, 2,827 non-falls) across three acquisition groups (Bed, Chair, Stand) and four sessions. An earlier draft was evaluated on a faulty archive extraction in which clips appeared three to four times and 728 duplicates straddled the train/test boundary, inflating every headline number;[1] all results here use a strict feature-group twin-free protocol: clips are grouped by feature fingerprint, one representative kept per group (5,572 representatives; 221 duplicate members removed; no conflicting-label groups), then stratified 64/16/20 into 3,565/892/1,115 train/validation/test (574 test falls, 541 non-falls). All model selection uses validation only; thresholds maximise validation F1; AUROC and threshold metrics carry 95% bootstrap intervals from saved score vectors (1,000 resamples, fixed seed); count metrics use Wilson intervals; FP/FN are deterministic counts; the five pre-specified paired-bootstrap comparisons against DTS+HGB remain significant after Bonferroni correction. Baselines: 2-layer LSTM (216,193 parameters), 2-layer GRU, temporal Transformer, Compact ST-GCN (244,161), FullST-GCN-COCO (1,107,905) trained from scratch (Yan, Xiong, and Lin, 2018), MiniROCKET via aeon (Dempster, Schmidt, and Webb, 2021; Middlehurst et al., 2024), MultiROCKET (Tan et al., 2022) (49,728 features; ridge with validation-selected regularisation), a four-block dilated causal TCN (Bai, Kolter, and Koltun, 2018) (43,713 parameters; JAX (Bradbury et al., 2018); Adam, gradient clipping, validation-AUROC checkpointing, 20 epochs), and DTS with LR/ExtraTrees/RF/HGB heads (grids in Appendix Table 6). MultiROCKET and the TCN consume sequences rebuilt from the raw archives via the released manifest; the rebuild reproduces DTS+HGB's AUROC exactly (0.9895) and MiniROCKET's within its published interval (0.9797), so the added rows are directly comparable. InceptionTime (Ismail Fawaz et al., 2020) ships as a configured script.

Main result and significance. Table 2: DTS+HGB attains AUROC 0.9895 [0.9851, 0.9935] and F1 0.951 [0.938, 0.963], best of thirteen models. The five pre-specified paired-bootstrap comparisons

[1]The earlier draft reported held-out AUROC 0.9998 and recall 1.000 for DTS+HGB; the strict protocol yields 0.9895 and 0.969. The corrected numbers supersede all previously circulated results.

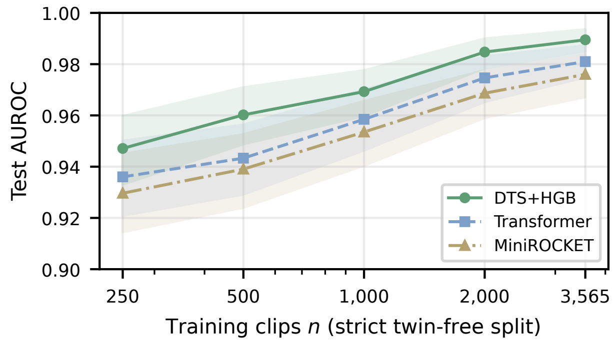

Figure 1: Sample efficiency on the strict twin-free split (shaded bands: 95% bootstrap intervals). DTS+HGB leads at every training size, with the largest margins at the smallest n.

all favour DTS+HGB significantly: +0.0183 [0.0105, 0.0266] vs GRU, +0.0085 [0.0024, 0.0151] vs Transformer, +0.0135 [0.0058, 0.0220] vs MiniROCKET, +0.0710 [0.0568, 0.0879] vs Compact ST-GCN, +0.0242 [0.0166, 0.0338] vs FullST-GCN-COCO. Among the added baselines, DTS+HGB significantly outperforms the TCN (+0.0265 [0.0163, 0.0377]) and *matches* the strongest baseline, MultiROCKET (0.9868 [0.9808, 0.9920]; paired margin +0.0027 [-0.0035, 0.0086], not significant), with 128 interpretable features versus 49,728 and fewer missed falls at the F1 point (18 vs 32). At matched MiniROCKET false positives (40), DTS+HGB detects 557 of 574 falls; at matched recall (548/574), it needs 34 false positives to MiniROCKET's 47.

Operating points: recall at fixed FPR. Deployment hinges on recall at acceptable alarm rates, since alarm fatigue, not ranking, is the binding constraint in continuous monitoring. DTS+HGB dominates MiniROCKET at every constraint: recall 0.796 [0.761, 0.827] vs 0.629 [0.589, 0.667] at 1% FPR, 0.862 vs 0.777 at 2%, 0.941 [0.918, 0.957] vs 0.861 [0.830, 0.887] at 5%, and 0.981 vs 0.970 at 10%. Against MultiROCKET the picture is honest and mixed: MultiROCKET recalls more falls at the strictest budgets (0.864 vs 0.796 at 1%; 0.882 vs 0.862 at 2%), the methods tie at 5% (0.944 vs 0.941), and DTS+HGB leads at 10% (0.981 vs 0.977) and at the F1 operating point (18 vs 32 missed falls). Operating-point choice between the two is thus a deployment decision at statistically indistinguishable AUROC; what the signature adds at parity is interpretability, a 388$\times$ smaller representation, and sub-3 ms CPU inference.

Sample efficiency. Retraining the three strongest families on stratified subsamples ($n \in \{250, \ldots, 3{,}565\}$; validation/test fixed) shows DTS+HGB ahead at every size: 0.9471 vs 0.9360 (Transformer) and 0.9296 (MiniROCKET) at n=250, with the largest margins in the scarcest regimes (Figure 1). This sweep, not the capacity heuristic, is the main support for the data-efficiency claim.

Distribution shift: sessions, fall origins, external datasets, subjects. Leave-session-out (subject proxy; FallVision publishes no subject IDs) gives mean AUROC 0.9649 over four folds (Appendix Table 4). Leave-fall-origin gives mean 0.8961: Chair 0.9844 and Stand 0.9916 transfer readily while Bed (0.7123) is the one hard fold, as the QDA analysis predicts (semi-reclined falls shift $\hat{\mu}_f, \hat{\sigma}_f$ outside the Chair+Stand support); the transfer gap is $\Gamma_{F1} = 0.951 - 0.880 = 0.071$ (Appendix Figure 5). Zero-shot transfer of the archived FallVision-trained model: URFD (Kwolek and Kepski, 2014) AUROC 0.9200 [0.8483, 0.9917] with recall 1.000 [0.886, 1.000] (30/30 falls) but 29 false positives, a score-calibration shift rather than a ranking failure (precision 0.508 [0.384, 0.632]; we do not estimate a calibrated threshold without a URFD validation split); GMDCSA-24 (Alam et al., 2024), an uncontrolled-home, low-resolution-webcam dataset, AUROC 0.9633 [0.9362, 0.9849] with the default threshold already usable (recall 0.924, precision 0.880, 10 FP, 6 FN on 160 clips: 79 falls, 81 ADL in our parse). Under *true* leave-one-subject-out on GMDCSA-24 (published subject IDs; four subjects), DTS+HGB attains mean AUROC 0.941 versus MultiROCKET 0.934, MiniROCKET 0.918, and α-only 0.917 (Table 3); the DTS$-$MultiROCKET margin (paired pooled out-of-fold AUROC $+0.012$ [-0.034, 0.056]) is not significant at four subjects, so we state the result as a match:

Model	S1	S2	S3	S4	Mean
DTS+HGB	0.953	0.969	0.970	0.871	**0.941**
MultiROCKET	0.988	0.882	0.948	0.918	0.934
MiniROCKET	0.988	0.816	0.942	0.926	0.918
α-only+LR	0.977	0.944	0.965	0.782	0.917

Table 3: True leave-one-subject-out AUROC on GMDCSA-24 (4 subjects). DTS+HGB matches the modern MultiROCKET baseline under person-level shift, with a small, non-significant mean advantage and zero sequence-learning parameters.

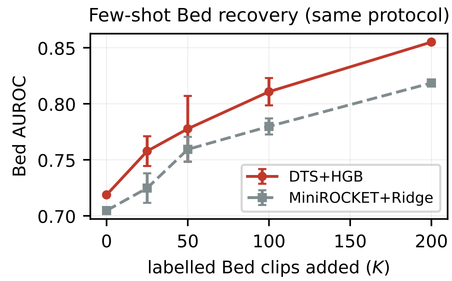

Figure 2: Few-shot Bed recovery under the identical protocol for both models (mean$\pm$95% interval over five seeds; fixed 1,547-clip Bed test). DTS+HGB recovers faster and higher than MiniROCKET+ridge at every adaptation budget.

a fixed 128-dimensional signature with zero sequence-learning parameters matches a state-of-the-art convolutional model under person-level shift.

Failure analysis and few-shot recovery. The Bed fold is characterised rather than hidden: it is a representation shift, not miscalibration (AUROC is threshold-free and re-tuning the Bed threshold barely moves F1). Missed Bed falls have higher hip-speed asymmetry ($\alpha \approx 0.33$) than detected ones ($\alpha \approx 0.21$), exactly the direction the QDA model predicts for semi-reclined, less front-loaded descents. Adding K labelled Bed clips to training lifts Bed AUROC monotonically from 0.714 (K=0) to 0.784 (K=50) and 0.848 (K=200), reproduced by an independent five-seed re-implementation (Appendix Figure 6; the recovery protocol reserves part of the Bed fold as the adaptation pool, so its zero-shot start differs slightly from the full-fold 0.7123); the fall/non-fall score separation widens from 0.45 to 0.59. Under the identical protocol (same base, adaptation pool, and fixed 1,547-clip Bed test; five seeds), DTS+HGB dominates a MiniROCKET+ridge baseline at every adaptation budget, from 0.719 vs 0.705 zero-shot to 0.855 vs 0.819 at K=200 (Figure 2): the structured representation also adapts more sample-efficiently. The hard fold reveals the limits of the inductive bias; the failure is interpretable, predicted by the theory, and largely closed with minimal target supervision. Figure 3 visualises the calibration story from the saved score vectors: FallVision and GMDCSA-24 are cleanly separated at the default threshold (non-fall mass above 0.5: 7.2% and 12.3%), while the Bed fold's high-score non-fall mass is partially cleared by adaptation; URFD vectors were not retained from the archived run, so its calibration shift is quantified in the text only.

Ablations and feature analysis. Single-family removal changes AUROC by at most -0.0039 (Appendix Table 7): the signature is intentionally over-complete, with partially redundant families providing robustness to missing or noisy keypoints. Zeroing all eight α coordinates costs only -0.0023 AUROC on real data; this does not show α is unnecessary, because other operators (slopes, velocities, accelerations) encode correlated order information, and the controlled benchmark, where marginal information is identical by construction, isolates the pure ordering contribution that the real-data ablation cannot (order-invariant operator subsets drop to chance there, Appendix Table 8). DTS-Net attention correlates positively but non-significantly with tree importances ($r = 0.650$, $p = 0.081$, $k = 8$ families) and is treated as exploratory; across all 128 coordinates, tree-importance ranks correlate with label-association ranks ($\rho = 0.523$ and $\rho = 0.704$ against Spearman and Welch statistics), and seven of the top twelve features are asymmetry coordinates. Extraction plus inference run in under 3 ms per clip on commodity CPU (<50 MB), excluding pose estimation.

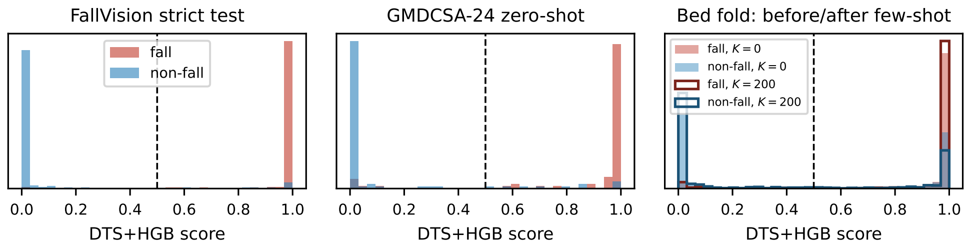

Figure 3: Class-conditional DTS+HGB score distributions from the saved score vectors (dashed line: fixed threshold 0.5). Left: FallVision strict test. Centre: GMDCSA-24 zero-shot (default threshold usable). Right: the hard Bed fold before ($K=0$, filled) and after ($K=200$, outlined) few-shot adaptation, which thins the high-score non-fall mass and widens the fall/non-fall separation from 0.44 to 0.59.

6 When should physics-guided temporal signatures work?

The theory and the two experimental regimes support an explicit scope statement. PGTS should help when: (1) the event mechanism is known, so primitives can be named a priori; (2) the discriminative signal is low-dimensional relative to the raw sequence; (3) labelled data are scarce relative to the capacity of competitive sequence models; (4) temporal order carries necessary information (Theorem 2 fails for order-invariant maps precisely then); and (5) domain shift acts interpretably through primitive distributions, as in the Bed fold, enabling diagnosis and few-shot repair. PGTS should be expected to fail or lose when: event structure is unknown or heterogeneous; many classes differ subtly; appearance or context cues dominate kinematics; abundant labelled data or pretrained models restore the advantage of learned representations; or the operator bank is misaligned with event timing, the failure mode realised in the benchmark when a fixed τ violates $\tau \leq p_1$ and the SNR sweep approaches 1. These conditions are testable in advance of training, which is what makes the recipe useful rather than merely retrospective.

7 Limitations

Real-data evidence is strongest for skeleton kinematics and binary event detection; we make no claim for large multi-class recognition. FallVision lacks public subject IDs (leave-session-out is a proxy; the leakage erratum and corrected protocol are described in Section 5); GMDCSA-24 LOSO has only four subjects, so the MultiROCKET comparison is underpowered and is stated as a match. PoseC3D and pretrained or fine-tuned skeleton-action systems are not evaluated; pretraining reintroduces large external data and may dominate, so evaluated-baseline superiority is the only claim. Bed-origin falls remain hardest (AUROC 0.7123 zero-shot) and recovery requires labelled target clips; URFD transfer requires threshold calibration. Subjects are healthy adults aged 22–27; elderly and frail-adult falls may violate the three-phase pattern more often. PGTS assumes known event mechanics and a low-dimensional discriminative signal; the finite-sample theorem idealises increments as independent and bounded. Runtime excludes pose estimation; DTS-Net attention is directional evidence, not validated explanation. Tree ensembles may be less natural than neural models for some end-to-end deployments.

8 Conclusion

When event dynamics are known and data are scarce, a fixed physics-guided temporal signature with a lightweight classifier is a theoretically grounded, empirically strong, and interpretable alternative to generic sequence learning. The impossibility and finite-sample theorems delimit exactly when temporal order must be modelled and when a single fixed statistic provably suffices; the controlled benchmark family realises both predictions; and the fall-detection case study shows the recipe surviving strict leakage-proof evaluation, low-data regimes, fixed-false-positive-rate operation, and person-level distribution shift against strong learned baselines. We release all artefacts for one-command reproduction.

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A Proofs

Theorem 2. By order-invariance, $\phi(\sigma(x)) = \phi(x)$ for every $\sigma \in S_T$ and every x. A negative is $X^- = \sigma(X^+)$ with $X^+ \sim P_+$ and σ independent of X^+. Pointwise, $g(X^-) = h(\phi(\sigma(X^+))) = h(\phi(X^+)) = g(X^+)$, so the class-conditional score laws coincide. For any threshold t, $\mathrm{TPR}(t) = \mathrm{FPR}(t)$, so the ROC curve is the diagonal, $\mathrm{AUROC} = \frac{1}{2}$, and the Bayes classifier on ϕ cannot beat the prior. The argument is pointwise in σ, so it covers deterministic permutations such as exact time reversal. $\square$

Theorem 3. Write $m = \lfloor \tau n \rfloor$, $S_\tau = \sum_{t \le m} w_t$, $S = \sum_{t \le n} w_t$, $\alpha = S_\tau / S$ (the stabiliser ϵ only decreases α and is absorbed). All $w_t \in [0, B]$. Fix $\varepsilon = \bar{\mu}\Delta/12 \le \bar{\mu}/2$.

Event side. Since $\tau \le p_1$, the first m increments have mean μ_s, so $E[S_\tau] \le \tau n \mu_s$ and $E[S] = n\bar{\mu}$. By Hoeffding (Hoeffding, 1963), $P(S_\tau \ge n(\tau\mu_s + \varepsilon)) \le \exp(-2\varepsilon^2 n^2/(mB^2)) \le \exp(-2\varepsilon^2 n/B^2)$ and $P(S \le n(\bar{\mu} - \varepsilon)) \le \exp(-2\varepsilon^2 n/B^2)$. On the complement, using $(\bar{\mu} - \varepsilon)^{-1} \le (1 + 2\varepsilon/\bar{\mu})/\bar{\mu}$ for $\varepsilon \le \bar{\mu}/2$,

$$\alpha_{\text{event}} \le \frac{\tau\mu_s + \varepsilon}{\bar{\mu} - \varepsilon} \le a_f + \frac{2\varepsilon\tau}{\bar{\mu}} + \frac{2\varepsilon}{\bar{\mu}} \le a_f + \frac{4\varepsilon}{\bar{\mu}} = a_f + \frac{\Delta}{3} < \theta.$$

Negative side. Conditional on the realised multiset $\{w_t\}$, the first m entries of the permuted sequence are a uniform sample without replacement, so by the Hoeffding–Serfling inequality (Hoeffding, 1963; Serfling, 1974), $P(|S_\tau^{\text{neg}} - \frac{m}{n}S| \ge \varepsilon n \mid \{w_t\}) \le 2\exp(-2\varepsilon^2 n^2/(mB^2)) \le 2\exp(-2\varepsilon^2 n/B^2)$. Combined with the lower-tail bound on S and $m/n \ge \tau - 1/n$, on the complement,

$$\alpha_{\text{neg}} \ge \frac{m}{n} - \frac{\varepsilon n}{S} \ge \tau - \frac{1}{n} - \frac{2\varepsilon}{\bar{\mu}} \ge \tau - \frac{\Delta}{6} - \frac{\Delta}{6} = \tau - \frac{\Delta}{3} > \theta,$$

using $n \ge 6/\Delta$ and $\tau - \Delta/3 > a_f + \Delta/2 \iff \Delta/6 > 0$. Summing the five failure probabilities with $\varepsilon = \bar{\mu}\Delta/12$ gives (3); the AUROC and balanced-error consequences follow as in the main text. Corollary 1 is Hoeffding's inequality applied to the k vote indicators. $\square$

Numerical verification. With μ_s=0.1, μ_d=1.0, p_1=0.3, p_2=0.5, τ=0.25, increments uniform on $[0, 2\mu]$ (B=2), 20,000 trials per n: the empirical left side of (3) is 0.192 (n=60), 0.100 (n=120), 0.030 (n=250), 0.003 (n=500), 0.000 (n=1000), never violating the bound, with empirical AUROC 0.976, 0.999, 1.000, 1.000, 1.000 respectively (`verify_theorem.py`).

B Capacity heuristic (intuition only)

Take $d_{\text{DTS}} = 128$ and, as a crude capacity proxy, $d_{\text{LSTM}} = 216{,}193$ (trainable parameters of the 2-layer LSTM baseline). The scale $n^* = d_{\text{LSTM}}/\ln d_{\text{LSTM}} \approx 17{,}599$ marks where crude $\sqrt{d/n}$ arguments favour the low-dimensional representation; the curves are parallel on log-log axes and never cross (Figure 4), the derivation omits constants, and FallVision's $n_{\text{train}} = 3{,}565$ is $4.9\times$ below n^*. This is intuition, not a bound; the learning curves in Figure 1 are the evidence.

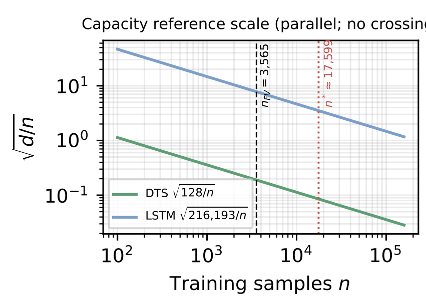

Figure 4: Heuristic capacity comparison: parallel $\sqrt{d/n}$ curves; n^* (dotted) is a derived scale, not a crossing point; $n_{\text{train}} = 3{,}565$ (dashed).

Session	n	Falls	F1	AUROC	Recall
1	1,513	679	0.866 [0.847, 0.884]	0.9382 [0.9259, 0.9492]	0.944
2	1,593	1,072	0.966 [0.958, 0.973]	0.9926 [0.9898, 0.9951]	0.954
3	2,064	903	0.925 [0.912, 0.937]	0.9840 [0.9797, 0.9879]	0.925
4	402	213	0.868 [0.836, 0.900]	0.9448 [0.9227, 0.9645]	0.953
Mean	–	–	0.906	0.9649	–

Table 4: Leave-session-out (subject proxy) for DTS+HGB, fixed threshold 0.5; 95% bootstrap intervals.

Held-out	n	Falls	F1	AUROC
Bed	1,747	922	0.768 [0.749, 0.788]	0.7123 [0.6868, 0.7357]
Chair	1,878	961	0.934 [0.923, 0.944]	0.9844 [0.9799, 0.9883]
Stand	1,947	984	0.937 [0.926, 0.949]	0.9916 [0.9886, 0.9945]
Mean	–	–	0.880	0.8961

Table 5: Leave-fall-origin robustness for DTS+HGB, fixed threshold 0.5; 95% bootstrap intervals.

C Additional real-data tables

D Original Bed-recovery reproduction

E Controlled-benchmark details and sweeps

Construction defaults: $T{=}150$, eight informative channels, drop SNR 8, onset $p_1{=}0.5$ (fall) / 0.4 (sit-to-stand), duration 0.2 (fall) / 0.3 (sit-to-stand), 500 held-out test pairs per run; τ selected on training data only by the summed-Welch rule; bag-of-frames uses per-channel value quantiles (order-invariant by construction). All runs, seeds, and per-run results are in `results_all.jsonl`; `synth_bench.py` regenerates everything and `analyze_bench.py` regenerates the tables.

F Reproducibility

Code, extracted features, split manifests, the deduplication audit, saved score vectors, hyperparameter grids, selected configurations, per-model thresholds, a numbers audit covering every headline value (`NUMBERS_AUDIT.md`), and seeded one-command reproduction scripts (`synth_bench.py`, `analyze_bench.py`, `verify_theorem.py`; `REPRO_COMMANDS.md`) are released with this paper.

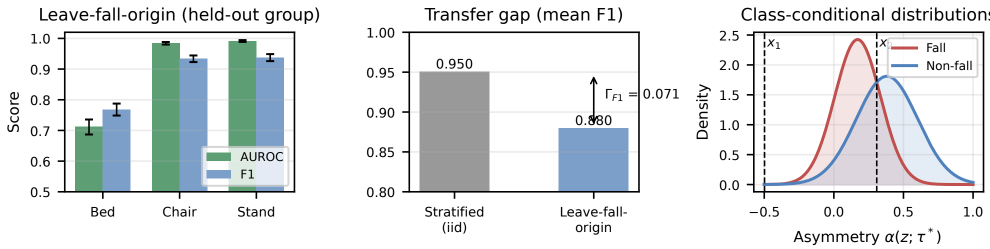

Figure 5: Left: leave-fall-origin AUROC and F1 with 95% intervals. Centre: transfer gap versus the stratified iid reference (reference shown to two decimals in the panel; $\Gamma_{F1} = 0.951 - 0.880 = 0.071$). Right: class-conditional hip-speed asymmetry distributions with QDA boundaries; only $x_2 \approx 0.308$ is active since $\alpha \geq 0$.

Model	Search space	Selected
DTS+LR	$C \in \{0.05, 0.1, 0.5, 1, 2, 10\}$	C=0.05
DTS+ET	trees $\{100, 300, 600\}\times$ feat. $\{\sqrt{d}, 0.25\}$	100 trees, max_features=0.25
DTS+RF	trees $\{100, 300, 600\}\times$ feat. $\{\sqrt{d}, 0.25\}$	600 trees, max_features=0.25
DTS+HGB	iter. $\{100, 300, 600\}\times$ lr $\{0.05, 0.1, 0.2\}$ $\times$ $\lambda_2\{0, 0.1\}$	600 iter., lr=0.1, λ_2=0.1
MiniROCKET	kernels $\{1000, 5000, 10000\}$	10,000

Table 6: Validation-only hyperparameter grids and selected configurations.

Configuration	F1	AUROC	ΔF1	ΔAUC
Full model	0.951	0.9895	–	–
−BBox-W	0.942	0.9882	−0.009	−0.0013
−Hip-Y	0.939	0.9855	−0.012	−0.0039
−Torso-Ang	0.945	0.9892	−0.006	−0.0003
−Hip-Spd	0.949	0.9897	−0.002	+0.0002
−Ctr-Spd	0.949	0.9904	−0.002	+0.0009
−Hip-Acc	0.947	0.9894	−0.004	−0.0001
−Shldr-Y	0.943	0.9882	−0.008	−0.0012
−Head-Y	0.951	0.9903	−0.001	+0.0008
Asymmetry zeroed (α=0)	0.939	0.9872	−0.012	−0.0023

Table 7: DTS+HGB feature ablation, main protocol (each row retrained; threshold re-tuned on validation).

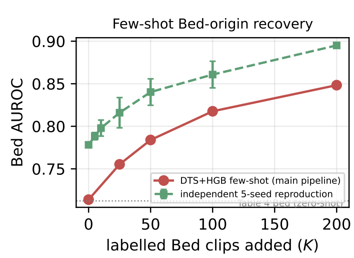

Figure 6: Original few-shot Bed-origin recovery: main-pipeline curve and an independent five-seed re-implementation (95% intervals); both recover the fold monotonically. The matched-protocol baseline comparison is in main-text Figure 2.

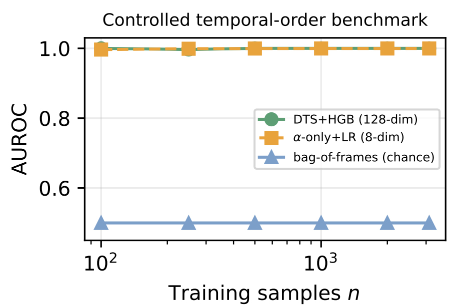

Figure 7: Original single-construction benchmark: structured signature and α-only saturate; order-invariant bag-of-frames sits exactly at chance for every n (Theorem 2).

Setting	DTS+HGB	α-only+LR	Bag-of-frames	MiniROCKET
Drop SNR μ_d/σ_s (fall, n=250)				
SNR 1	1.000	0.879±0.004	0.500	1.000
SNR 2	1.000	0.993±0.001	0.500	1.000
SNR 4	1.000	1.000	0.500	1.000
SNR 8	1.000	1.000	0.500	1.000
SNR 16	1.000	1.000	0.500	1.000
Event onset p_1 (fall, n=250)				
p_1=0.15	1.000	1.000	0.500	–
p_1=0.3	1.000	1.000	0.500	–
p_1=0.5	1.000	1.000	0.500	–
p_1=0.7	1.000	1.000	0.500	–
α-only with misspecified fixed τ=0.30 vs train-selected τ				
p_1=0.15	selected: 1.000		fixed 0.30: 1.000	
p_1=0.3	selected: 1.000		fixed 0.30: 0.993±0.001	
p_1=0.5	selected: 1.000		fixed 0.30: 1.000	
p_1=0.7	selected: 1.000		fixed 0.30: 0.999	
Distractor channels added (fall, n=250)				
+0 noise ch.	1.000	1.000	0.500	1.000
+8 noise ch.	1.000	1.000	0.500	1.000
+32 noise ch.	1.000	1.000	0.500	1.000
+120 noise ch.	1.000	1.000	0.500	1.000
Event duration p_2-p_1 (fall, n=250)				
dur 0.05	1.000	0.992±0.001	0.500	
dur 0.1	1.000	0.999	0.500	
dur 0.2	1.000	1.000	0.500	
dur 0.4	1.000	1.000	0.500	
Sequence length T (fall, n=250)				
T=30	1.000	0.999	0.500	
T=60	1.000	1.000	0.500	
T=150	1.000	1.000	0.500	
T=300	1.000	1.000	0.500	
Operator-bank ablation, DTS+HGB (fall, n=250)				
full 16-operator bank			1.000	
− all α			1.000	
α only			1.000	
order-sensitive ops only			1.000	
order-invariant ops only			0.500	
quantiles/min/max only			0.500	
slopes only			1.000	
endpoints only			0.997±0.002	

Table 8: Controlled-benchmark sweeps (held-out AUROC, mean±sd over three seeds, 500 test pairs). Bag-of-frames remains at exactly 0.500 in every setting, as Theorem 2 requires, and order-invariant operator subsets of the signature itself (quantiles, min/max) also sit exactly at chance, confirming that the order-sensitive operators carry all discriminative signal. α-only degrades as drop SNR approaches 1 and dips mildly under a misspecified fixed τ or very short events, while the full operator bank and train-selected τ remain at ceiling. At these settings every order-sensitive method solves the constructions, including with 120 distractor channels: the family measures the necessity of order sensitivity, not separation among order-sensitive methods.